

Test 4

April 30, 2024

Problem 1

Solve: $\sqrt{2x+7} - 5 = 0$

Problem 2

Solve: $\sqrt{11x-18} - x = 0$

Problem 3

Solve: $4x^3 - 2 = 498$

Problem 4

Solve: $n^2 + 4n = 21$

Problem 5

Solve: $4x^2 - 14 = 2$

Problem 6

Find a quadratic equation with 3 and -1 as its solutions.

Problem 7

Solve: $y^4 - 9y^2 = -20$

Problem 8

If the side of a square is increased by 4 the area is multiplied by 9. Find the side of the original square.

Problem 9

Jack can wash and wax the family car in one hour less than Bob can. The two working together can complete the job in 40 min. How much time would each require if they worked alone?

Problem 10

Solve: $xy = 72$ and $(x + 2)(y - 4) = 128$

Problem 11

A merchant bought a number of barrels of apples for \$200. He kept two barrels and sold the remainder at a profit of \$4 per barrel making a total profit of \$52. How many barrels did he originally buy?

Problem 12

Let $g(x) = 4x^2 - 4$.

Find $g(0)$.

Problem 13

Let $w(n) = 2^{n+2}$.

Find $w(3)$.

Problem 14

Let $f(a) = 3a + 4$ and $g(a) = a^3 + 2a$.

Find $f(3) + g(3)$

Problem 15

Let $h(t) = t + 5$ and $g(t) = 5t - 5$.

Find $(h \cdot g)(5)$.

Problem 16

Let $f(a) = 2a - 4$ and $g(a) = a^2 + 2a$.

Find $(f \circ g)(-4)$.

Problem 17

Find the inverse of $g(x) = \frac{3}{x-4}$.

Problem 18

Solve: $2^{2x} = \frac{1}{64}$.

Problem 19

Solve: $\log_8 x = 2$

Problem 20

Solve: $\log_3 81 = x$

Extra Credit

\$5000 is invested in an account earning 11.5% interest compounded monthly for a 20 year period. What is the balance at the end of 20 years?